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A REPORT ON MAXIMUM POWER POINT TRACKING
SYSTEM

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ABSTRACT

This project attempts to investigate and analyse two different MPPT approaches. The goal of maximum power point tracking algorithms, sometimes known as MPPT, is to devise methods for extracting the greatest power from a PV source. The following MPPT commands are investigated and reviewed in this project: Perturb and observe, and Incremental Conductance while familiarising oneself with the simulation platform.

Along with this, the necessity of renewable energy and its various sources is highlighted, as well as a thorough examination of solar power.

Modelling the MPPT system, analysing and interpreting simulation results, and demonstrating their positive behaviour are all done in the MATLAB Simulink environment.

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1. INTRODUCTION

1.1 Purpose, Scope and Limitation

According to the International Energy Agency (IEA), worldwide energy consumption has increased by 70%, well above expected projections. This analysis also found that the trend of rising energy consumption will continue for many more decades, as economic growth and expansion continue unabated.

Each year, energy consumption grows by at least 2%, indicating that it will continue to rise.

Fossil fuels provide roughly 90% of global energy needs, therefore increased use will undoubtedly put more strain on already depleted fossil fuel sources.

The pattern of energy usage implies that per capita energy use in developed countries is already disturbingly high. The rate of increase in energy consumption in developing countries is presently enormous. Because the developing world accounts for 80% of global population, a massive increase in energy consumption per capita might spell disaster for key fossil fuel reserves such as coal, petroleum, and natural gas.

According to recent forecasts, developing countries' energy consumption will increase by 40% in the next ten years. Rapid population growth, major industrial expansion, urbanisation, and rising per capita income will all contribute to an increase in energy demand.

Climate change and greenhouse gas emissions will result from this growth in energy consumption.

The way we utilise energy is rapidly changing. To keep global temperatures from rising, the transition to renewable energy sources must happen faster, not just in electricity generation but also in heating, construction, and transportation.

By 2050, renewables could provide four-fifths of the world's electricity, drastically reducing carbon emissions and aiding in climate change mitigation. However, solar and wind power must be properly integrated, with sustainable biofuels playing an important role as well.

As a result, considering the demand for renewable energy, the report's goal is to research and assess ways for harnessing clean solar energy to its highest efficiency.

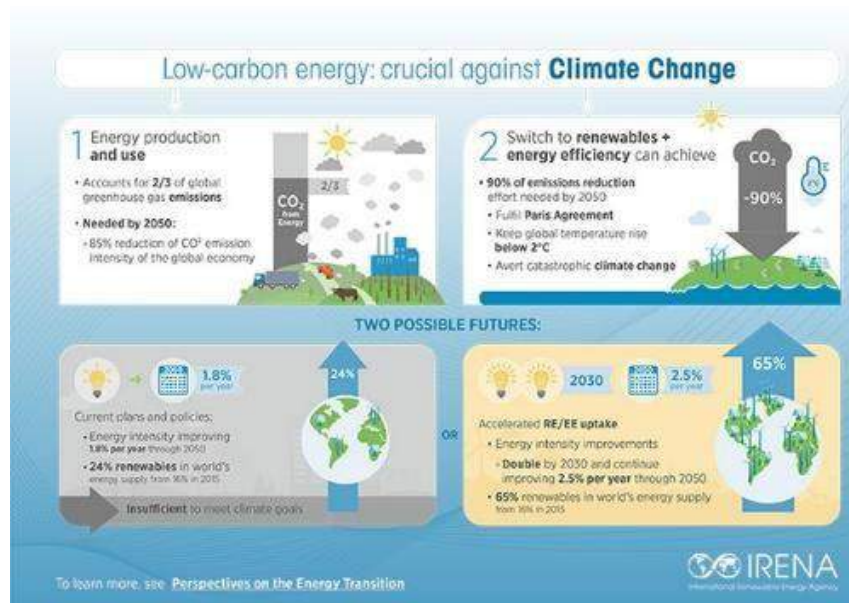


Figure 1.1: Low-carbon energy

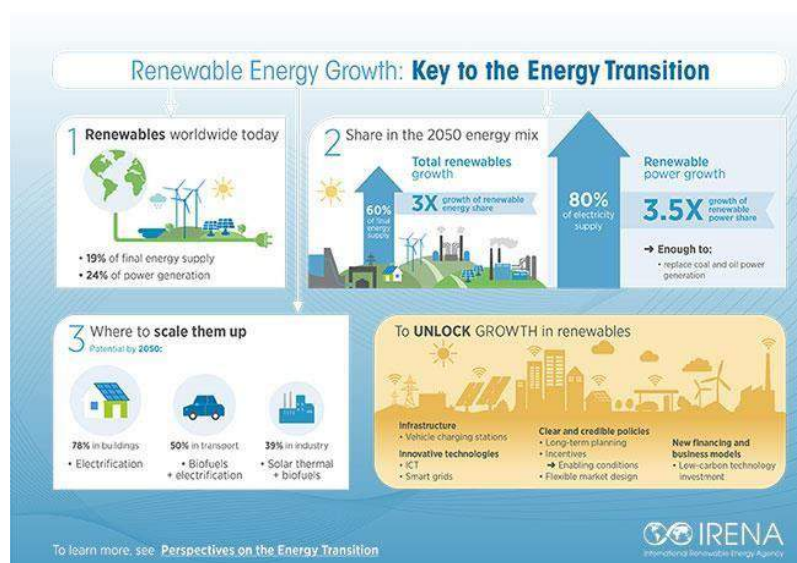


Figure 1.2: Renewable energy growth

1.2 Sources of Renewable Energy

1.2.1 Wind Power

Wind turbines allow us to harness the wind's energy-generating potential. The turbine's blades spin clockwise when the wind blows, capturing energy. This causes the wind turbine's primary shaft, which is coupled to a gearbox within the nacelle, to spin. The gearbox converts the wind energy into electricity and transmits it to the generator. The electricity is then sent to a transformer, which adjusts the voltage levels to match the grid.

Under the correct conditions, a mid-ranged household turbine of 5 kW may provide roughly 8,000 kWh to 9,000 kWh of electricity per year. Smaller turbines with a power output of roughly 2 kW can generate up to 3,000 kWh of electricity. Larger residential turbines have the potential to reach 15,000 kWh.

1.2.2 Hydropower

During the hydrological cycle, the first phase in the creation of energy in a hydroelectric plant is the gathering of run-offs from seasonal rain and snow in lakes, streams, and rivers. The run-off is channelled into dams further downstream. The water flows via a dam, into the hydropower plant, and through a turbine, which is a big wheel. The generator is driven by the turbine, which turns the energy of falling water into mechanical energy. After this procedure, the water is discharged back into the lakes, streams, or rivers, and power is transported to the villages via transmission lines. Hydropower accounts for 16% of global electricity generation.

1.2.3 Biomass

Photosynthesis is the process by which plants harness the sun's energy. These plants release the trapped energy when they burn. In

this way, biomass acts as a natural battery, storing solar energy and releasing it as needed.

1.2.4 Geothermal

The thermal energy that is generated and stored within the layers of the Earth is known as geothermal energy. The gradient so formed results in continual heat conduction from the earth's core to the surface. This gradient can be used to heat water to produce superheated steam, which can then be used to power steam turbines. The fundamental disadvantage of geothermal energy is that it is mainly restricted to areas near tectonic plate boundaries, though recent developments have led to its spread.

1.2.5 Solar Power

Solar power is the electricity produced by the Sun's solar energy. Although solar power accounts for a modest percentage of worldwide energy output, it is used extensively in some countries. It has two applications: Solar-thermal power plants employ a variety of methods to generate electricity from the sun. The sun's energy is used to boil water, which then powers a steam turbine, which generates electricity in a similar way to coal or nuclear power plants. Solar panels, also known as photovoltaic cells, are the most common way to harness solar energy. They are stacked practically edge-to-edge in solar power facilities to capture sunlight in huge fields.

Semiconductor materials are used to make the cells. The sun's rays free electrons from their atoms when they strike the cells. This permits electrons to flow freely across the cell, resulting in the generation of electricity. A power electronic circuit must be constructed to extract maximum power from the photovoltaic integrated system since solar photovoltaic cells have non-linear V-I characteristics that are depending on insolation and environmental factors. This project creates a circuit model and uses MATLAB/Simulink to simulate it.

1.3 Source and Methods

Research materials from several journals and a NEPTEL lecture video on Design of Photovoltaic Systems were utilised to understand and apply the concept.

The circuit model is designed in MATLAB/Simulink, and to become familiar with the programme, MathWorks learning tools and other online tutorials are employed.

1.4 Report Organization

2. LITERATURE SURVEY

According to studies, a solar panel transforms between 21 and 40 percent of the energy incident on it into electrical energy. To improve the efficiency of the solar panel, a Maximum Power Point Tracking algorithm is required.

Perturb and Observe (hill climbing method), Incremental conductance, Fractional Short Circuit Current, Fractional Open Circuit Voltage, Fuzzy Control, Neural Network Control, and more MPPT approaches exist. Perturb and observe (P&O) and incremental conductance are the most often utilised approaches due to their simplicity in design, shorter time to track the MPP, and other cost considerations.

When the meteorological circumstances (irradiance level) change abruptly, P&O interprets the change in MPP as a change in MPP due to disturbance rather than irradiance, and sometimes calculates the erroneous MPP.

The Incremental Conductance technique, on the other hand, avoids this difficulty because it calculates MPP using two voltage and current samples. However, instead of being more efficient, the algorithm's complexity is much larger than the prior one, increasing the cost of implementation. As a result, we must strike a balance between intricacy and efficiency.

It may be noticed that we are acquiring which type of converter to acquire best efficiency. We have chosen a boost converter since it provides greater voltage at the output than it does at the input. We have the option of using a buck-boost converter, but we have chosen a boost converter due to our need for simplicity. It is easy to set up and use, and it works well in both stationery and time-varying environmental situations.

In this project, we will try to put these two strategies into practise while also examining the theoretical notion and qualities presented in other works.

3. PHOTOVOLTIC SYSTEM MODELLING

3.1 Historical Perspective of Solar Cell

A photovoltaic (PV) cell, commonly known as a solar cell, is a type of electrical component that produces electricity when exposed to photons, or light particles. The photovoltaic effect, discovered in 1839 by French physicist Edmond Becquerel, is the term applied to this conversion.

It was not until the 1960s that photovoltaic cells found their first practical application in satellite technology. Rooftop solar panels, which are made up of PV cell modules, first appeared around the end of the 1980s. Since the beginning of the twenty-first century, photovoltaic capacity has continuously increased, thanks to the construction of massive solar farms.

3.2 PV Module, Panel and Array

A single solar PV cell produces very little energy, ranging from 0.1 to 2 watts. However, using such a low-power unit as a system building block is impractical. As a result, the required number of such cells are assembled to make a feasible commercially available solar unit known as a solar module or PV module.

To generate larger voltages, currents, and power levels, photovoltaic cells are electrically coupled in series and/or parallel circuits. Photovoltaic modules are the core building elements of PV systems, consisting of PV cell circuits sealed in an environmentally protective laminate.

Photovoltaic panels are made up of one or more PV modules that are pre-wired and field installable. unit.

A photovoltaic array is the complete power-generating unit, consisting of any number of PV modules and panels.

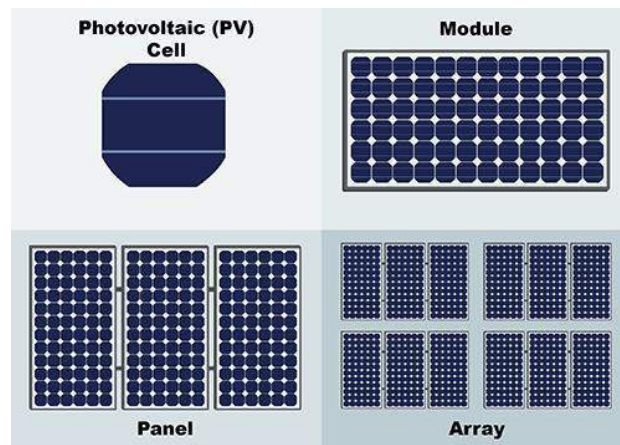


Figure 3.1: PV Module, Panel and Array

3.3 Working of a PV Cell

A solar cell is basically a junction diode, although its construction is a little bit different from conventional p-n junction diodes. A very thin layer of p-type semiconductor is grown on a relatively thicker n-type semiconductor. We then apply a few finer electrodes on the top of the p-type semiconductor layer.

These electrodes do not obstruct light to reach the thin p-type layer. Just below the p-type layer there is a p-n junction. We also provide a current collecting electrode at the bottom of the n-type layer. We encapsulate the entire assembly by thin glass to protect the solar cell from any mechanical shock.

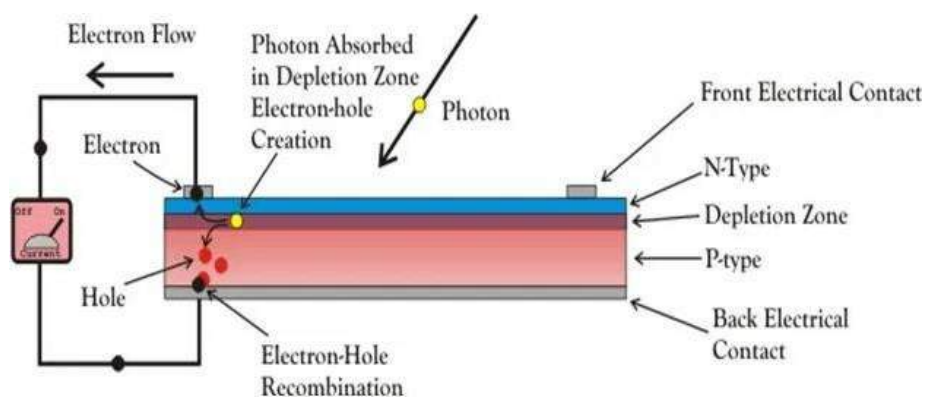


Figure 3.2: Working of a PV cell

When irradiance hits the surface of solar PV cell, an electrical field is generated inside the cell. This process separates positive and negative charge carriers in an absorbing material (joining p-type and n-type). In the presence of an electric field, these charges can produce a current that can be used in an external circuit. This generated current depends on the intensity of the incident radiation.

3.4 Modelling of PhotoVoltic Cell

Whenever the sunlight falls on the PV materials, it generates electrons flowing in the external circuit known as photocurrent short circuit current and is calculated by equation (i).

In modelling of the PV cell, it is indicated by a current source as shown in Fig. 2. There appears a voltage at the output terminal if it is open circuited and called open circuit voltage V_{oc} calculated by equation (ii).

This voltage causes a current through the P-N junction just like a diode. This diode and the current source (I_{ph}) are put in parallel as depicted in Fig. 2. As the photogenerated current starts flowing, some of the electron-hole recombination occurs that reduces the originally generated electrons; this loss of current is presented by a shunt resistance (R_{sh}).

Series resistance (R_s) indicates the resistance faced by the current as it flows through the bulk material, external metal contacts and to the load. Manufacturers always manage to keep the effects of both these resistances as low as possible to improve the working of the PV module. Fig. 2 indicates the single diode model of the solar photo voltaic cell.

The photocurrent, also known as short-circuit current is generated when the solar rays strike the solar module and is calculated by Eq. (1)

$$I_{sc} = \frac{G}{1000} [I_{scr} + K_i(T_c - T_r)] \quad (1)$$

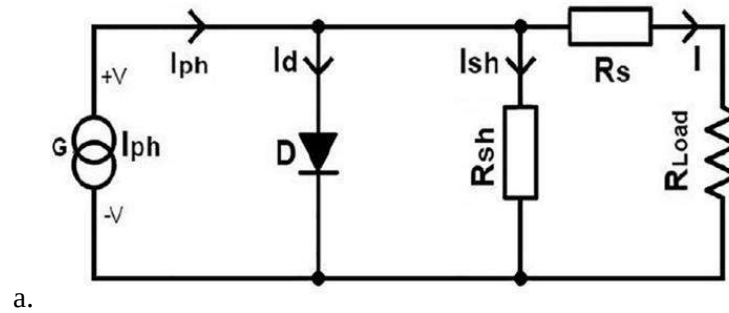
where G solar irradiance, I_{sc} photocurrent, I_{sc} reverse saturation current, K_i temperature coefficient, T_c cell temperature, T_r reference temperature
The open circuit voltage as generated at the output of the single diode model of the solar photovoltaic cell is calculated by Eq. (2)

$$V_{oc} = \ln \left(\frac{I_{sc}}{I_o} + 1 \right) \left(\frac{nkT_c}{q} \right) \quad (2)$$

where V_{oc} open circuit voltage, I_{sc} photo current, I_o saturation current, n ideality factor, k Boltzmann constant, T_c cell temperature, q Electron charge
Eq. (3) gives the relation between the voltage and current of the solar cell.

$$I = I_s \left[\exp \left(\frac{qV}{kT} \right) - 1 \right] - I_{ph} \quad (3)$$

where I_s saturation current, q Electron charge, V voltage across the diode, k Boltzman constant, T absolute temperature (K), I_{ph} light generated current.



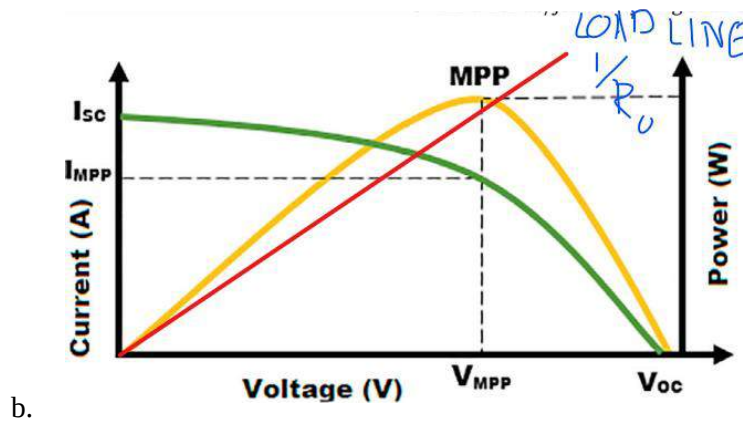


Figure 3.3: Model of a PV cell a. Equivalent circuit

b. I-V, P-V Characteristics

The characteristic curve closely and define two of the points. First one is the short circuit current I_{sc} and the second one is the open circuit voltage V_{oc} . Short circuit current is the current where the cell voltage is zero. Open circuit voltage is the voltage at which the cell current is zero. The point at which I_{mp} and V_{mp} meet is the maximum power point. This is the point at which maximum power is available from the PV cell. If the load line" crosses this point precisely, then the maximum power can be transferred to this load. The value of this load resistant would be given by:

$$R_{mp} = \frac{V_{mp}}{I_{mp}}$$

From equations 1 and 2 above we observe that I_{sc} is linearly proportional to the solar irradiation and V_{oc} is dependent on insolation but in a logarithmic fashion as shown in Figure 3.4 a.

Figure 3.4 b. represents the dependence of I_{sc} , V_{oc} and maximum operating point on temperature.

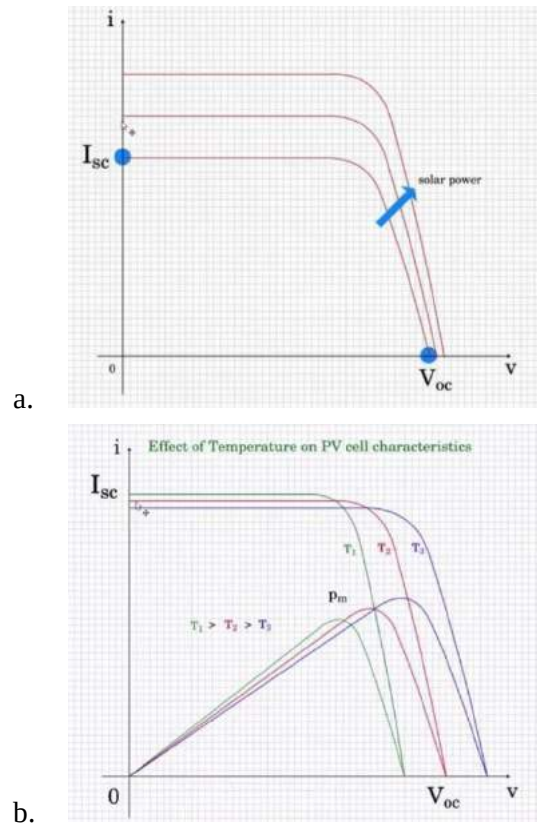


Figure 3.4: Dependence of PV cell on a- Irradiance b- Temperature

3.4.1 Datasheet Reading

The modelling of MPPT system depends on the parameters of the PV cell and therefore it is essential to understand the datasheet.

Figure 3.5 shows the datasheet of a Polycrystalline PV cell

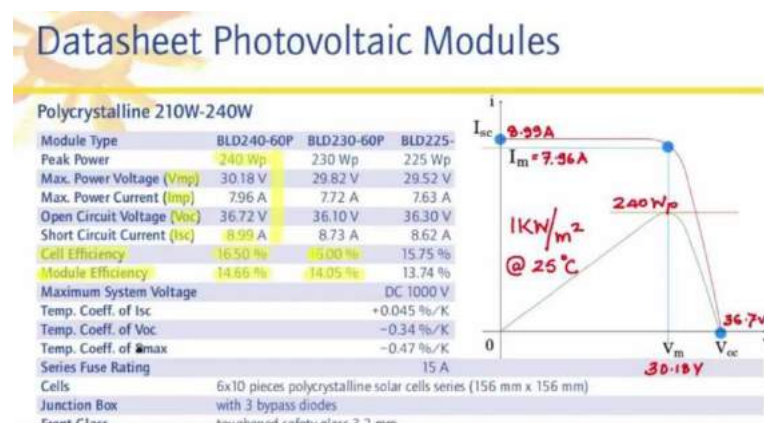


Figure 3.5: PV Datasheet

3.4.2 Cell Efficiency

The efficiency is the most used parameter to compare the performance of one solar cell to another. Efficiency is defined as the ratio of energy output from the solar cell to input energy from the sun. In addition to reflecting the performance of the solar cell itself, the efficiency depends on the spectrum and intensity of the incident sunlight and the temperature of the solar cell. Therefore, conditions under which efficiency is measured must be carefully controlled to compare the performance of one device to another.

$$P_{\max} = V_{OC} I_{SC} FF$$

$$\eta = \frac{V_{OC} I_{SC} FF}{P_{in}}$$

$$P_{in} = L \cdot A_{cell}$$

Where- I_{sc} = short-circuit current

V_{oc} = open circuit voltage

FF = fill factor

η = efficiency

L = insolation

A PV module consists of area that does not capture any solar irradiation and thus efficiency of module is always less than the efficiency of cell as shown in the datasheet above.

3.4.3 Fill Factor

The short-circuit current and the open-circuit voltage are the maximum current and voltage respectively from a solar cell.

However, at both operating points, the power from the solar cell is zero. The "fill factor", more commonly known by its abbreviation

"FF", is a parameter which, in conjunction with V_{oc} and I_{sc} , determines the maximum power from a solar cell. The FF is defined as the ratio of the maximum power from the solar cell to the product of V_{oc} and I_{sc} so that:

$$FF = \frac{P_{mp}}{V_{oc} * I_{sc}} = \frac{V_{mp} * I_{mp}}{V_{oc} * I_{sc}}$$

Graphically, the FF is a measure of the "squareness" of the solar cell and is also the area of the largest rectangle which will fit in the IV curve. The FF is illustrated in Figure 3.6.

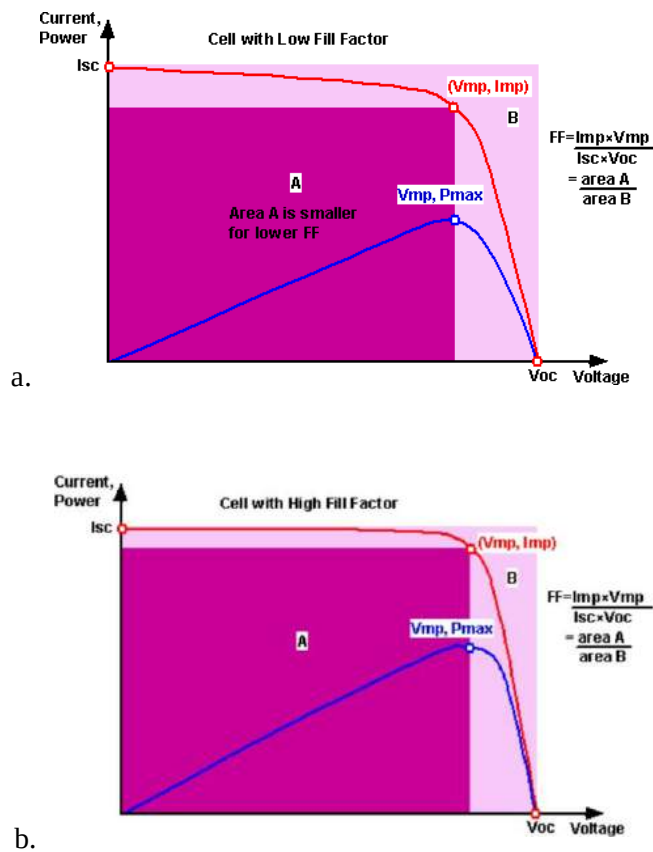


Figure 3.6 : Fill factor a. Low b. High

The quality of solar system depends upon the fill factor of solar panels. To find the quality of the solar panel fill factor is

used. A good panel has fill factor in the range of 0.7 to 0.8. for a bad panel it may be as low as 0.4.

3.5 Series and Parallel Combination of PV Cells

To boost the efficiency of a PV system, solar cells can be linked in series or parallel to form solar panels.

3.5.1 Cells in Series

When two identical cells are linked in series, the system's short circuit current remains the same, but the open circuit voltage is doubled. We can write the following relationships if the cells are identical.

$$\begin{aligned}I_{sc1} &= I_{sc2} = I_{sc} \\V_{oc1} + V_{oc2} &= 2V_{oc}\end{aligned}$$

When two dissimilar cells are connected in series, their open circuit voltages add up, but the net short circuit current is somewhere between I_{sc1} and I_{sc2} .

3.5.2 Cells in Parallel

When two identical cells are connected in parallel. The system's open circuit voltage would be the same as a single cell's open circuit voltage. However, the system's short circuit current would be twice that of a single cell. We can write the following relationships if the cells are identical:

$$\begin{aligned}I_{sc1} + I_{sc2} &= 2I_{sc} \\V_{oc1} &= V_{oc2} = V_{oc}\end{aligned}$$

Short circuit currents add up when two dissimilar cells are connected in parallel, while the open circuit voltage is between V_{oc1} and V_{oc2} .

4. DC -DC CONVERTERS

4.1 Need of Chopper Circuits with MPPT

DC/DC converters are employed in situations where an average output voltage that can be higher or lower than the input voltage is required. The selection of a suitable DC/DC converter for both the MPPT system and its integration into the facility array has a considerable impact on the photovoltaic system's optimal performance. The aim of a dc-to-dc converter, which connects the load and the PV module, is to transfer the maximum amount of power from the PV module to the load. The load impedance as seen by the source is adjusted and matched at the point of peak power with the source to transfer the greatest power by adjusting the duty cycle.

As a result, we will explore and evaluate several topologies of DC-DC converters in this chapter, as well as analyze what characteristics to consider while constructing a photovoltaic MPPT model.

4.2 Boost Converter

The fundamental circuit layout of a DC-DC boost converter circuit is shown in Figure, which includes a power switch (M), a diode (D), an inductor (L), a capacitor (C), a switching controller, and a load (R). This design can be used to connect a low-voltage PV array to a high-voltage battery bank input voltage or any DC load. The output voltage of the DC-DC boost converter will be boosted or stepped up to be greater than the input voltage. To increase the input voltage to the required output voltage, the controller will manage the switch for turn-on and turn-off. The diode will be in reversed bias when the switch is turned on, and electrical energy will be stored in the inductor.

As a result, current will be supplied to the load by the capacitor. The stored electrical energy in the inductor is transferred to the capacitor and load when the switch is turned off. Continuous-conduction mode (CCM)

and discontinuous-conduction mode (DCM) are the two modes of operation for DC-DC boost converters (DCM). The inductor current is always greater than zero when the DC-DC boost converter is operating in CCM, but it drops to zero after each switching cycle when it is operating in DCM.

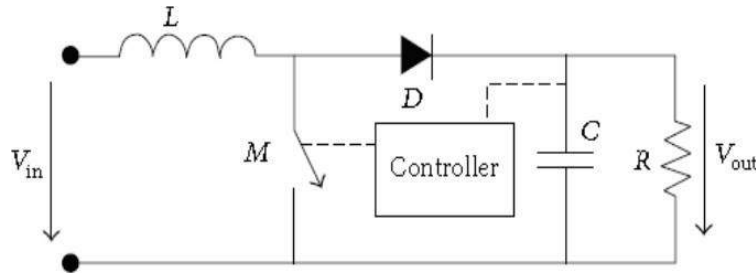


Figure 4.1: Boost converter circuit

4.3 Buck Converter

Figure shows the basic circuit topology of a DC-DC buck converter, which includes a power switch (M), a diode (D), an inductor (L), a capacitor (C), a switching controller, and a load. The DC-DC buck converter works as a step-down system, converting a high input voltage to a low output voltage with a magnitude that is always lower than the input voltage. By adding the LC low pass filter to the basic circuit of this converter, the goal of this circuit is to produce a totally DC output. From a high PV array voltage, this DC-DC buck converter can be connected to a low voltage DC load or battery bank.

Because of their simple topology, low control difficulties, low number of components, and lack of isolation, DC-DC buck converters are widely employed in very high range step down converters and low power range regulators. Most DC-DC buck converters are used in battery charging, controlling the high input voltage using PWM to provide the low output voltage required by the batteries, and MPP tracking to maximize the output power obtained from PV arrays. Standalone solar PV pumping system for water delivery in rural areas,

solar battery charger, MPPT tracking for grid-connected, and off-grid PV system are some of the solar PV applications that use a DC-DC buck converter.

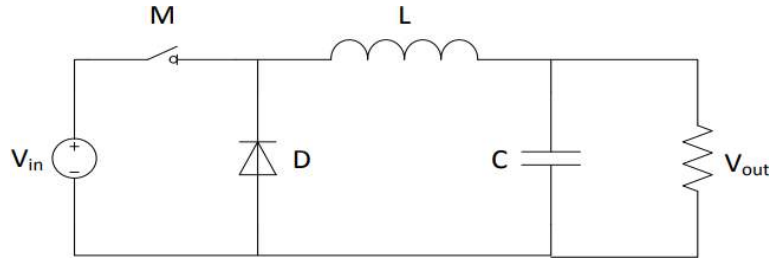


Figure 4.2: Buck Converter

4.4 Buck-Boost Converter

The structure of a DC-DC buck-boost converter is very similar to that of a DC-DC boost converter, with the exception that the switching element is placed before the inductor (L), as seen in Figure.

To connect with suitable PV array voltage, DC load, or battery input voltage, the DC-DC buck-boost converter, also known as a step-up/down and bidirectional converter, may generate either lower or higher output voltage from its input voltage. A cascaded connection of two basic converters, DC-DC buck and boost converters, results in a DC-DC buck-boost converter.

The duty cycle, D , of this converter can be changed to adjust the output. The converter will function in buck mode if the duty cycle is less than 50%, and the output voltage will be less than the input voltage. The converter will operate in boost mode when the duty cycle is greater than 50%, and the output voltage will be greater than the input voltage. The operating frequency of the inductor, the maximum current and voltage that the inductor can withstand, and the gate driver circuit that generates the PWM switching signals for triggering the power switch are all

design characteristics that should be addressed when constructing this converter.

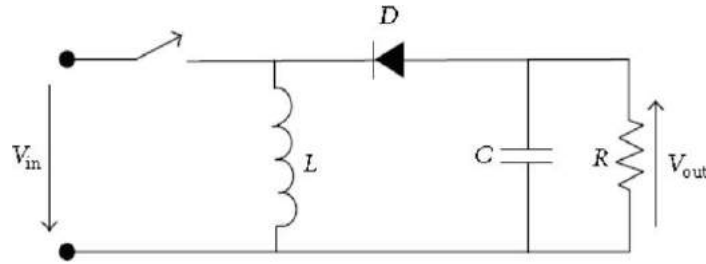


Figure 4.3: Buck-Boost converter

4.5 Comparison Between the Converters

The DC-DC buck, boost, and buck-boost converters are similar in that they use the same components in their circuits, but they differ in terms of individual characteristics and operating objectives. When comparing the performance of buck and boost converters, several parameters such as inductors, capacitors, and power switches with gate drivers can be taken into account. The DC-DC boost converter requires more inductance than the DC-DC buck converter, while the DC-DC buck converter requires a larger and more expensive capacitor to smooth the PV module input current.

Furthermore, while the DC-DC boost converter has a lower current rating for power switches and gate drivers, the DC-DC buck converter requires high-side power switches drivers, which are more complex and expensive to implement than the DC-DC boost converter. Furthermore, the DC-DC boost converter's existing free-wheeling diode can prevent reverse current, whereas the DC-DC buck converter requires an additional blocking diode to conduct PV maximum current. As a result of its improved dynamic performance and low cost of implementation, the DC-DC boost converter has more advantages.

The choice of DC-DC converter is determined by whether the PV arrays are connected in series or parallel. The DC-DC buck converter is better for PV arrays connected in series, but the DC-DC boost converter is better for PV arrays connected in parallel.

The shadowing effect may be the reason of the malfunctioning PV module, which reduces the output power generated by the solar PV system. As a result, the choice of DC-DC converter is critical for optimizing solar PV output power in partial shading due to the shadowing effect. The output current in a series connection is equal, however in a boost topology, the input current is greater than the output current, posing an operating limitation. As a result, both the current and the PV output power decline dramatically. Furthermore, in a parallel connection, the PV array output current equals the total of the individual PV currents. If any PV module experiences partial shade, the remaining PV modules will continue to operate normally, and the partial shading will have little impact on the total PV array output power. PV arrays connected in series with a DC-DC buck converter, on the other hand, have a free-wheeling diode that prevents partial shading and increases overall PV output power by avoiding malfunctioning PV modules.

5. MPPT MATLAB/ SIMULINK MODEL

5.1 Introduction to MPPT

For extracting maximum power from the solar panel and transferring maximum power from the PV module to the load, a maximum power point (MPPT) is utilized.

MPPT strategies are required to keep PV arrays working at its MPP. It addresses the limitations of high initial installation costs and low energy conversion efficiency for this operating condition. Consider the operation of a traditional (non-MPPT) charge controller to better understand how MPPT works. When a traditional controller is used to charge a drained battery, the modules are simply connected to the battery. This compels the modules to work at battery voltage, which is often not the optimal operating voltage for producing maximum available power.

The PV Module Power/Voltage/Current graph depicts a typical 75W module's traditional Current/Voltage curve under normal test settings of 25°C cell temperature and 1000W/m² of insolation. PV module power delivered vs. module voltage is also seen in this graph (fig.4). For the purposes of this example, the traditional controller simply connects the module to the battery, forcing it to function at 12V. The traditional controller artificially limits power production to 53W by forcing the 75W module to operate at 12V.

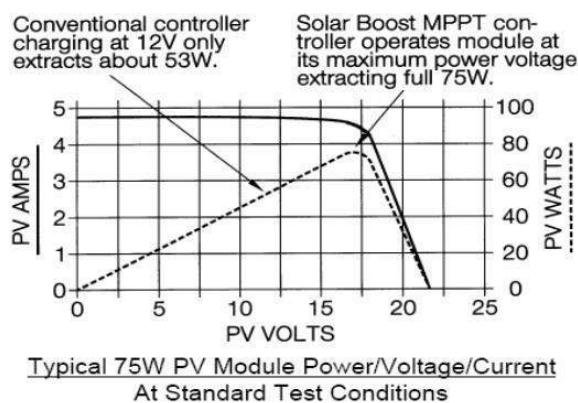


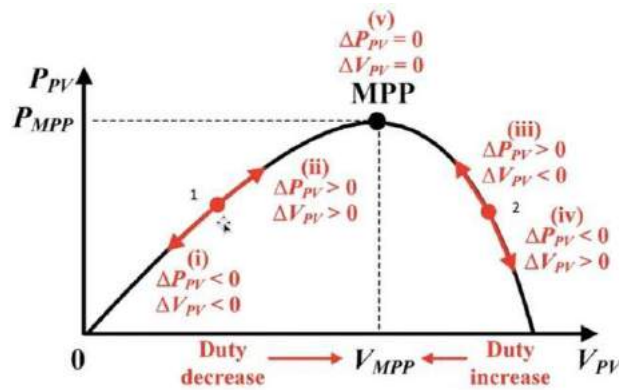
Figure 5.1: PV Controller without MPPT

5.2 Perturb and Observe Algorithm

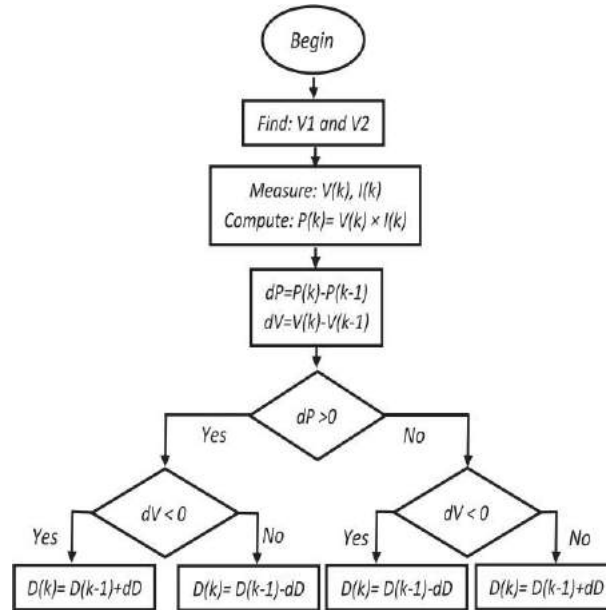
The Perturb & Observe algorithm has been widely utilized due to its ease of implementation, as seen in Figure 5.2 b's flowchart. The operating point converges at the MPP after a continuous process of observation and perturbation. The algorithm forecasts the time to approach MPP by comparing the power and voltages of time (K) with the sample at a time (K-1). If the power change is positive, a little voltage fluctuation affects the power of the solar panel, and the voltage perturbation continues in the same direction. However, if delta power is negative, the MPP is far away, and the perturbation must be reduced to approach it.

The P&O algorithm is summarized in Figure 5.2 c. As a result, the entire PV curve is examined by minor perturbations to discover the MPP, which increases the algorithm's response time.

When the perturbation size is increased, steady state oscillations around the MPP result.



a



b.

Scheme of the P&O algorithm.

Perturbation	Delta P	Resulting Perturbation
+ve	+ve	+ve
+ve	-ve	-ve
-ve	+ve	-ve
-ve	-ve	+ve

c.

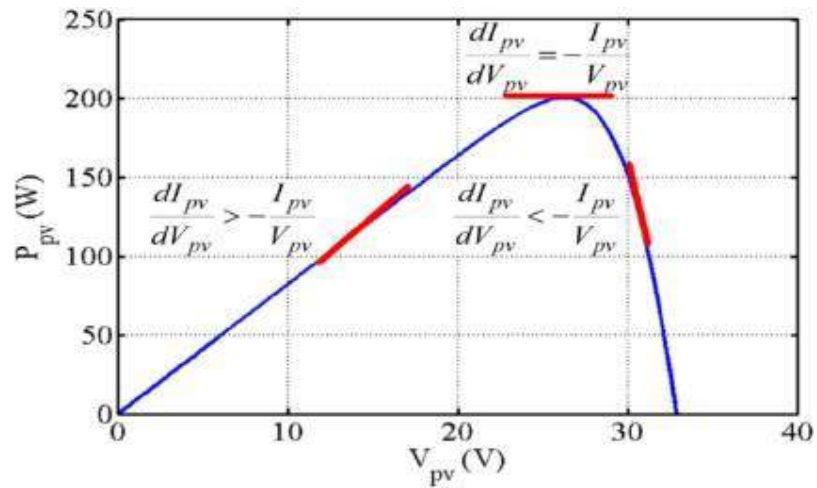
Figure 5.2: a- Concept, b.- Flowchart, c.- P&O table.

5.3 Incremental Conductance Algorithm

The IC approach overcomes the shortcoming of the perturb and observe method in tracking peak power under rapidly changing atmospheric conditions. The IC can determine when the MPPT has reached the MPP and when it is no longer necessary to disrupt the operational point. If this criterion is not met, the relationship between dI/dV and $-I/V$ can be used to compute the direction in which the MPPT operating point must be perturbed. The fact that dP/dV is negative when the MPPT is to the right of the MPP and positive when it is to the left of the MPP gives rise to this connection. This algorithm outperforms P&O in that it can tell when the MPPT has arrived at the MPP, whereas P&O oscillates about it. In addition,

In addition, incremental conductance is more accurate than perturb and observe at tracking quickly increasing and declining irradiance conditions. When compared to P&O, one disadvantage of this method is its increased complexity.

$$\begin{aligned} \frac{dP}{dV} &= 0 \text{ at MPP} \\ \frac{dP}{dV} &> 0 \text{ at the left of MPP} \\ \frac{dP}{dV} &< 0 \text{ at the right of MPP} \\ \frac{dP}{dV} &= \frac{d(VI)}{d(V)} = I + V \frac{dI}{dV} \end{aligned} \quad \begin{aligned} \frac{dI}{dV} &= -\frac{I}{V} \text{ at MPP} \\ \frac{dI}{dV} &> -\frac{I}{V} \text{ at the left of MPP} \\ \frac{dI}{dV} &< -\frac{I}{V} \text{ at the right of MPP} \end{aligned}$$



a.

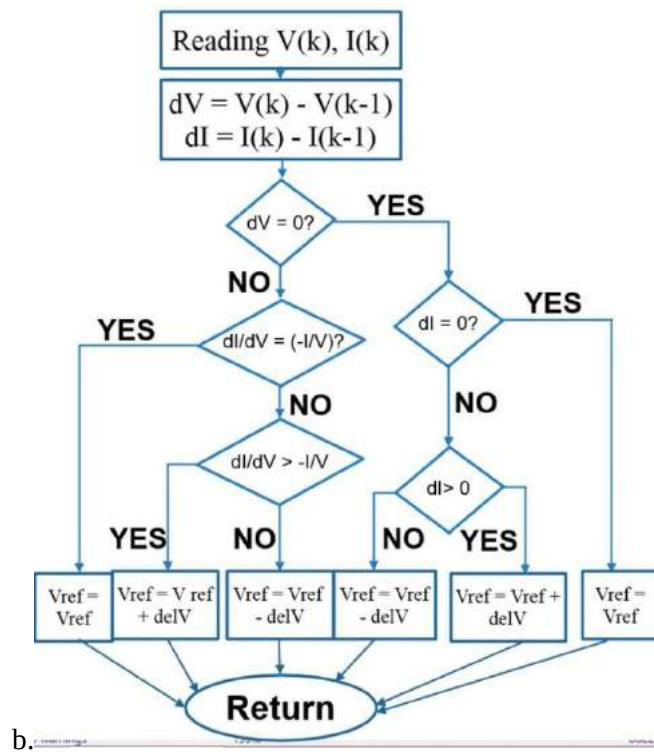


Figure 5.3: Incremental Conductance a- Concept b- Flowchart

5.4 Design of Simulink Model

A Simulink circuit model of a Photovoltaic MPPT system is shown in Figure 5.4. We will look at the functional blocks and subsystems that make up the technique in this section.

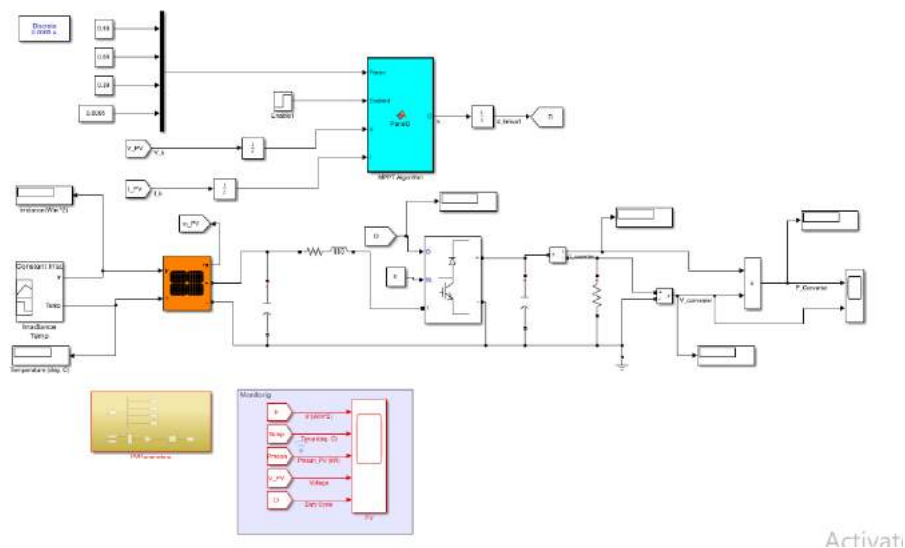
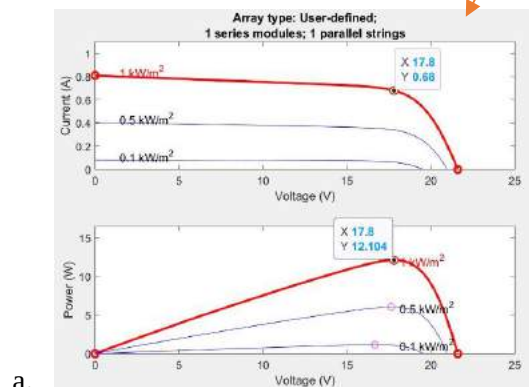
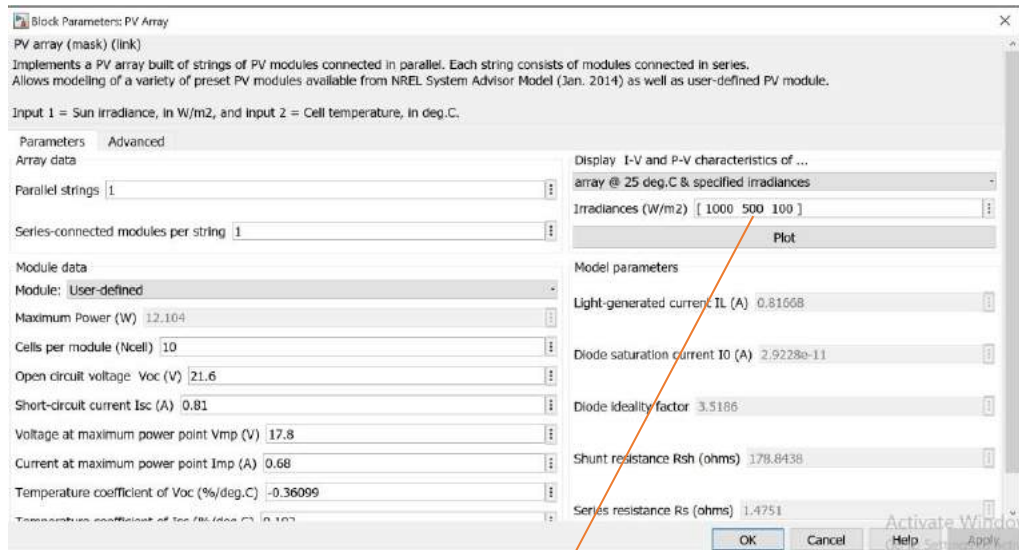


Figure 5.4: Overall Simulink Model

5.4.1 Solar Panel

The solar panel for the project is modelled using a PV array functional block from the Simscape library. When we double-click the block, a dialogue emerges that allows us to select an appropriate panel from a pre-defined list or create our own panel from the user-defined section, as seen in Figure 5.5.a. When you click Plot, MATLAB automatically draws the PV array's characteristics curve.



a.

Specification of solar module TTB12W.

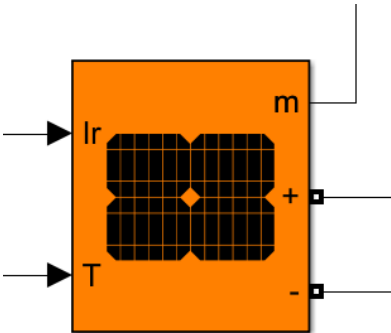
Characteristic	Value
Rated Power (P_{max})	12 W
Tolerance	$0 \pm 3\%$
Voltage Maximum Power (V_{max})	17.8 V
current at Maximum power (I_{max})	0.68A
voltage at Open circuit (V_{oc})	21.6 V
Photo Current (I_{sc})	0.81A

b.

Figure 5.5: a. Block Parameter window and Characteristic Curve for PV Array

b. Specification of Solar Module

Irradiance and temperature are input parameters for the PV array block, and a measuring port m is output. It also features a + and - terminal for connecting the circuit further.



a.

b.

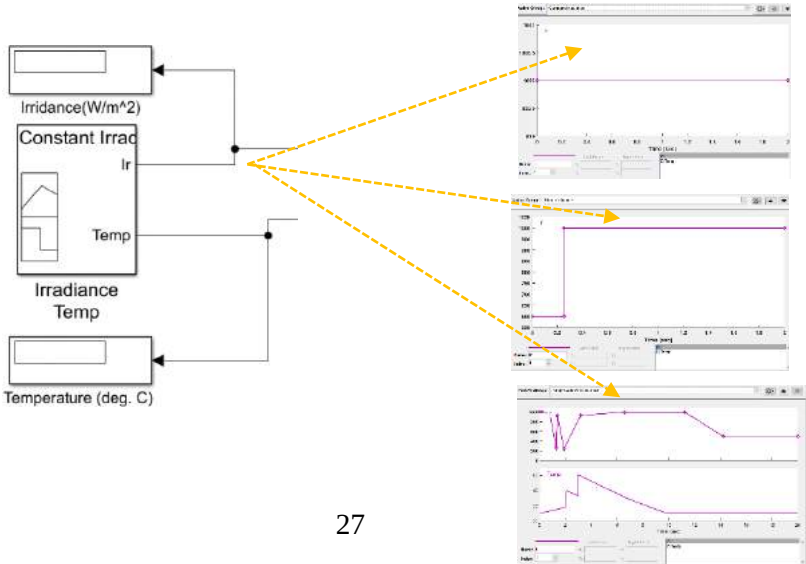
Measurements, returned as a five-element vector. You can select these signals using the Bus Selector block in the Simulink® library.

Signal	Signal Name	Definition
1	V_PV	PV array voltage (V)
2	I_PV	PV array current (A)
3	I_diode	diode current (A)
4	Irradiance (W/m2)	Irradiance (W/m2)
5	Temperature (deg C)	Temperature (degrees C)

Figure 5.6: a. PV array block b. Measurement parameters available at output port

To replicate real-world environmental circumstances, a signal building block is linked to the PV array's input ports. This block simulates three irradiance conditions: constant, step, and ramp-up/down.To read the irradiance and temperature data at any given time, two display blocks from the Simulink/ Sinks library are utilised.

a.



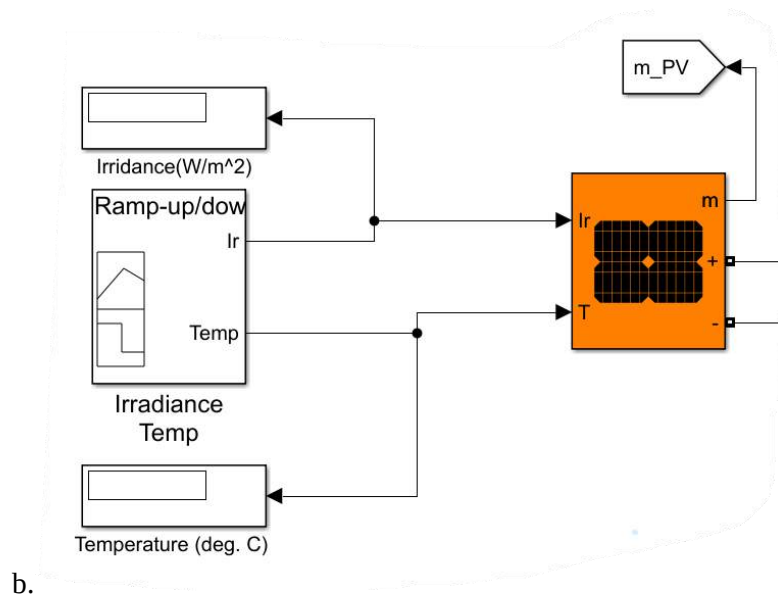


Figure 5.7: a. Signal Builder with three conditions b. Complete model of solar panel

5.4.2 Boost Converter

From the Simulink/Power electronics library, the boost converter block is chosen. A built-in (IGBT) and a diode make up the block. Depending on the load, converter parameters such as input capacitance, inductance, and output capacitance are considered.

The boost converter block is available in three different configurations, with the Average Model (D) providing the quickest simulation result.

The Figure 5.8 shows complete DC-DC interface circuit.

The duty cycle port is attached to a from block, the value of which is calculated using the MPPT algorithm.

To measure the circuit voltage and current, the Voltage and Current sensor block from the Sensor and Measurement library is used.

The power of the entire system is estimated using the measured data and a product block from the Math operation library. The output waveform and values are also visualized using a scope and display blocks from the sink library.

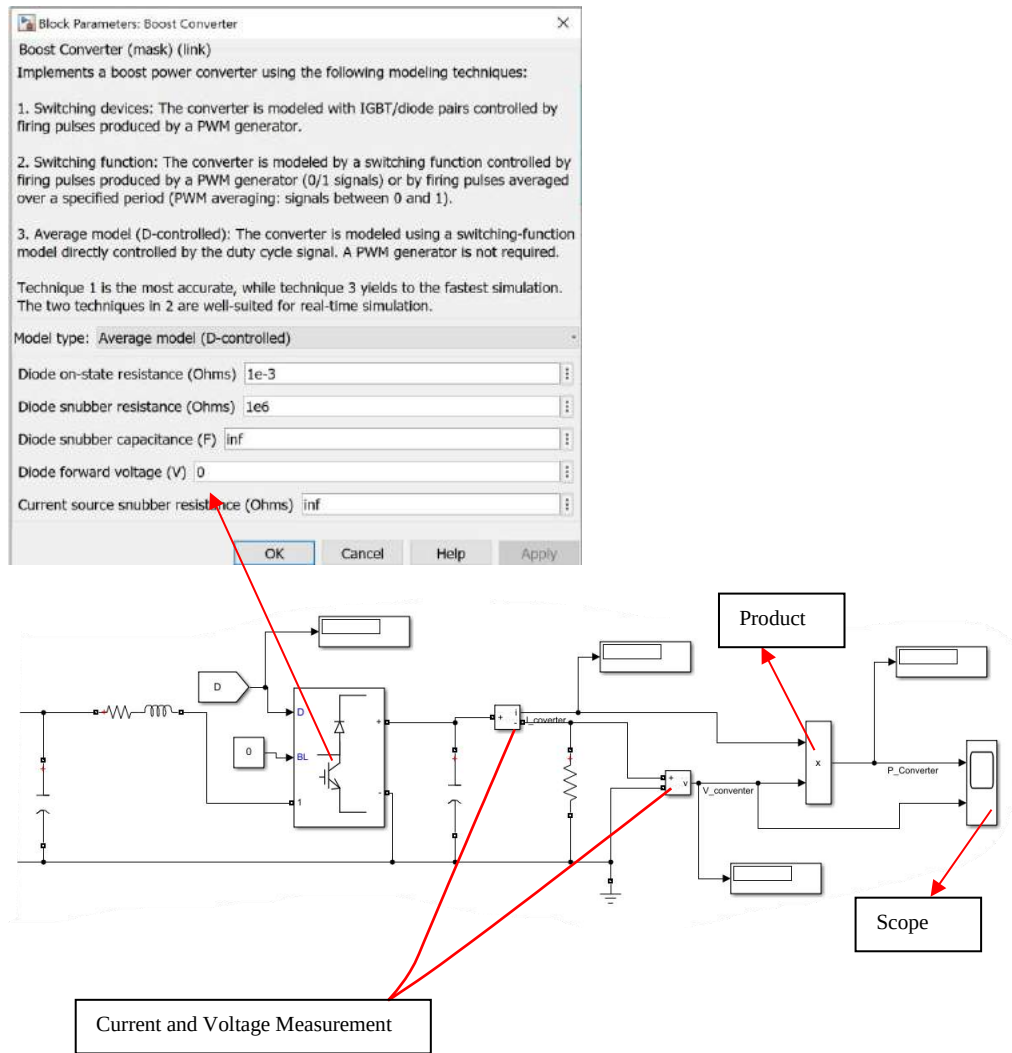


Figure 5.8: Boost Converter Model

5.4.3 MPPT Algorithm

The MPPT technique is based on the flowcharts mentioned earlier in the document. The algorithm is written in MATLAB using the HCL coder library's Function block. The input parameters are Param, Enabled, V, and I, while the functional block's output is duty cycle D.

Param provides four duty cycle parameters (initial, maximum, minimum, and step size), which are combined using a four channel Mux block from the Signal Routing library. The mux passes the input to the function as and when it is called in the code. Using a bus selector, the values of V and I are provided from the PV array's measurement port.

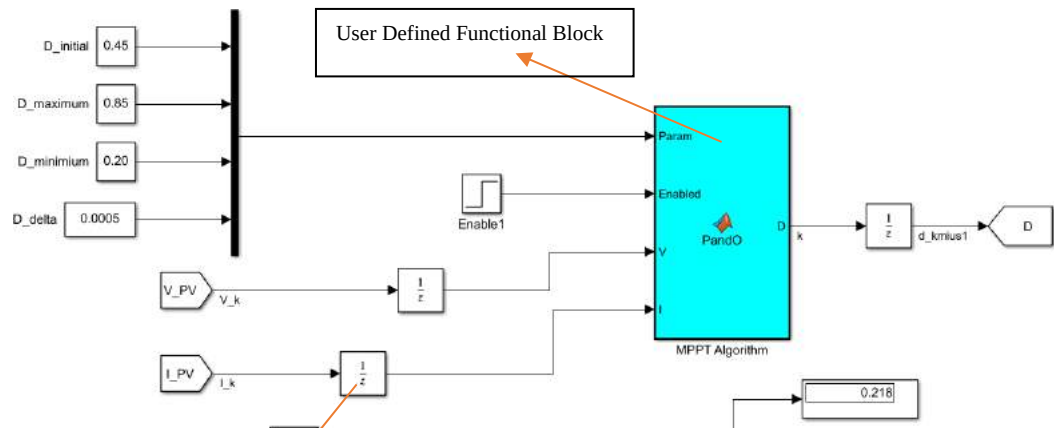


Figure 5.9: MPPT Algorithm

A unit delay block is placed before the input to obtain a value of voltage and current from a previous state as specified in the flowcharts and perturb and Observe concept.

Code For Perturb & Observe:

```
function D =PandO(Param, Enabled,V, I)

% MPPT controller based on the Perturb & Observe algorithm.
% D output = Duty cycle of the boost converter (value between 0 and 1)
%
% Enabled input = 1 to enable the MPPT controller
% V input = PV array terminal voltage (V)
% I input = PV array current (A)
%
% Param input:
Dinit = Param(1); %Initial value for D output
Dmax = Param(2); %Maximum value for D
Dmin = Param(3); %Minimum value for D
deltaD = Param(4); %Increment value used to increase/decrease the duty cycle D
% ( increasing D = decreasing V)

persistent Vold Pold Dold ;

dataType = 'double';
%Perturb and observed
if isempty(Vold)
    Vold=0;
    Pold=0;
    Dold=Dinit;
end
```

```

P= V*I;
dV= V - Vold;
dP= P - Pold;
if dP ~= 0 & Enabled ~=0
    if dP < 0
        if dV < 0
            D = Dold - deltaD;
        else
            D = Dold + deltaD;
        end
    else
        if dV < 0
            D = Dold + deltaD;
        else
            D = Dold - deltaD;
        end
    end
else
    D=Dold;
end

if D >= Dmax | D<= Dmin
    D=Dold;
end
Dold=D;
Vold=V;
Pold=P;

```

Code For Incremental Conductance:

```

function D =InC(Param, Enabled,V, I)
% MPPT controller based on the Perturb & Observe algorithm.
% D output = Duty cycle of the boost converter (value between 0 and 1)
%
% Enabled input = 1 to enable the MPPT controller
% V input = PV array terminal voltage (V)
% I input = PV array current (A)
%
% Param input:
Dinit = Param(1); %Initial value for D output
Dmax = Param(2); %Maximum value for D
Dmin = Param(3); %Minimum value for D
deltaD = Param(4); %Increment value used to increase/decrease the duty cycle D
% ( increasing D = decreasing V )

% Incremental Conductance Method
Vrefmax = 21.6;
Vrefmin = 0.0;
Vrefinit = 15;
deltaV = 0.5;
persistent Vold Dold Iold;
dataType = 'double';
if isempty(Vold)
    Vold = 0;
    Iold = 0;

```

```

    Dold = Dinit;
end
dV = V - Vold;
dI = I - Iold;
if (dV == 0 && Enabled ~= 0)
    if (dI == 0)
        D = Dold;
    else
        if (dI > 0)
            D = Dold - deltaD;
        else
            D = Dold + deltaD;
        end
    end
else
    if (dI/dV == -I/V)
        D = Dold;
    else
        if (dI/dV > -I/V)
            D = Dold - deltaD;
        else
            D = Dold + deltaD;
        end
    end
end

if (D >= Dmax || D <= Dmin)
    D = Dold;
end
Dold = D;
Vold = V;
Iold = I;

```

5.4.4 Power Block

The powergui block allows you to solve your circuit using one of the following methods:

- Simulink's variable-step solver is used in Continuous.
- Discretization of the electrical system to solve a problem in discrete time steps.
- The phasor solution can be continuous or discrete.

In addition, the powergui block provides access to tools for analyzing steady-state and simulation findings, as well as advanced parameter design.

To simulate any Simulink model including Simscape™ Electrical™ Specialized Power Systems blocks, you'll require the powergui block. It keeps track of the comparable Simulink circuit that embodies the model's state-space equations.

When one powergui block is used in a model:

- To get the best performance, put the powergui block in the top-level diagram.
- Make sure the block has the name powergui on it.

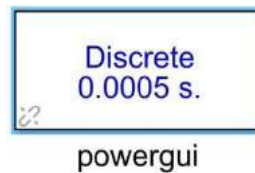


Figure 5.10: Power Block

5.4.5 Switching Between MPPT Techniques

We can develop a new algorithm that allows us to switch between strategies by incorporating a switch block into the design. For example, InC is said to be more efficient than P&O under fast-changing situations while also being more sophisticated.

We will not have to employ a sophisticated approach all the time because we will be able to activate it as needed with this method. This is accomplished using variant and triggered subsystems.

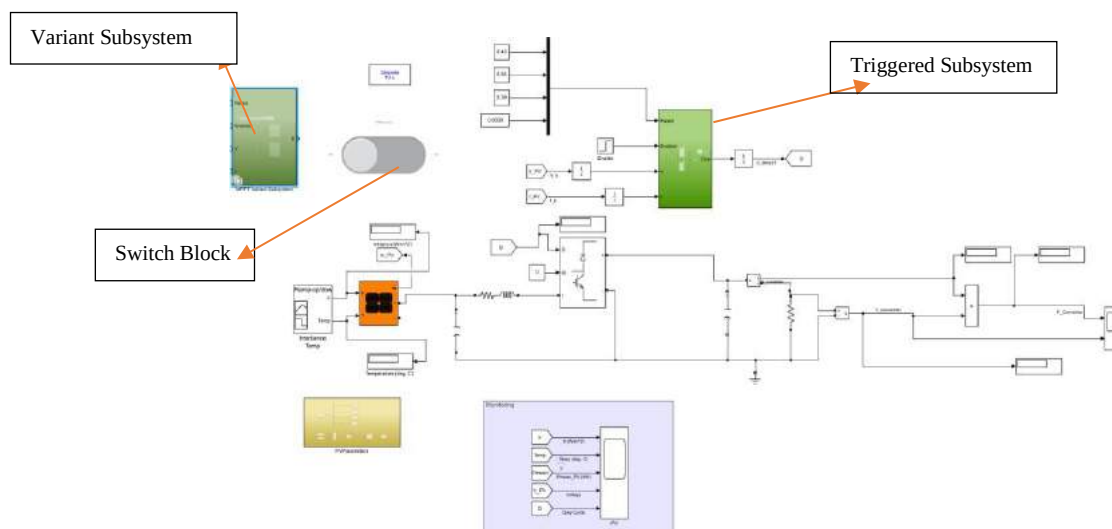


Figure 5.11: Modified Simulink model enabling us to switch between algorithms.

5.4.5.1 Variant Subsystem

It, too, is divided into two subsystems, one for each technique.

The input and output parameters are identical to those used in the original model.

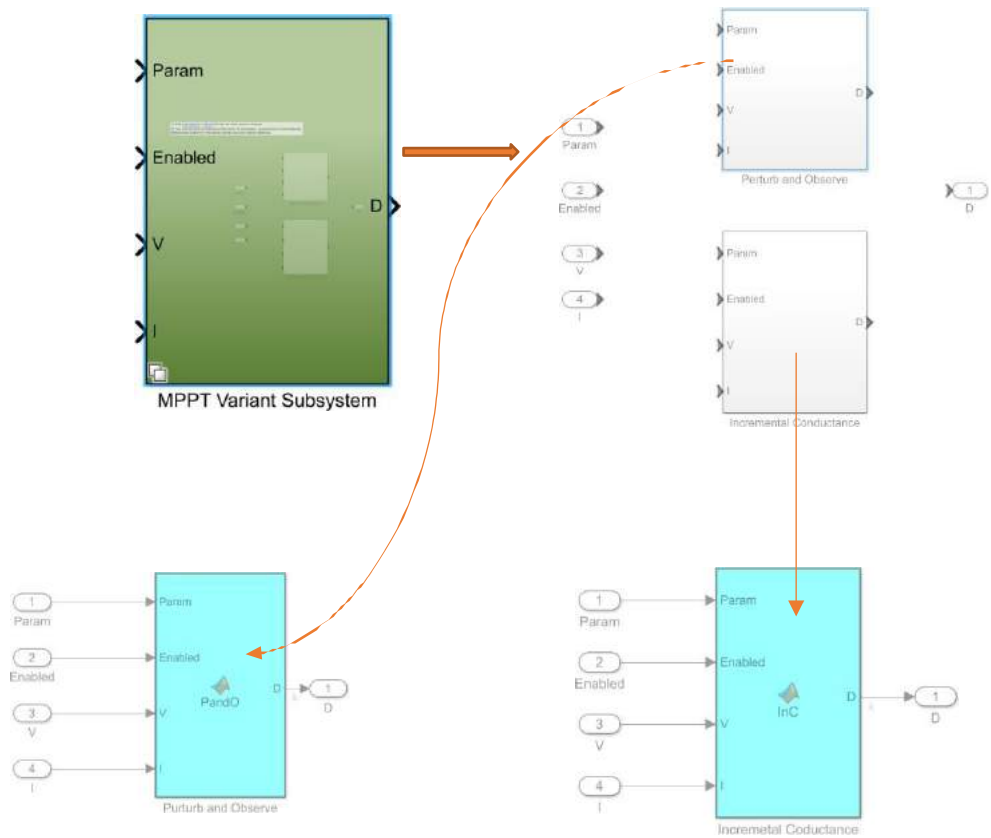


Figure 5.12: Variant subsystem Model

The variation manager specifies the conditions for modelling the variant subsystem, and if the conditions are met, one of them is chosen and the simulation is executed.

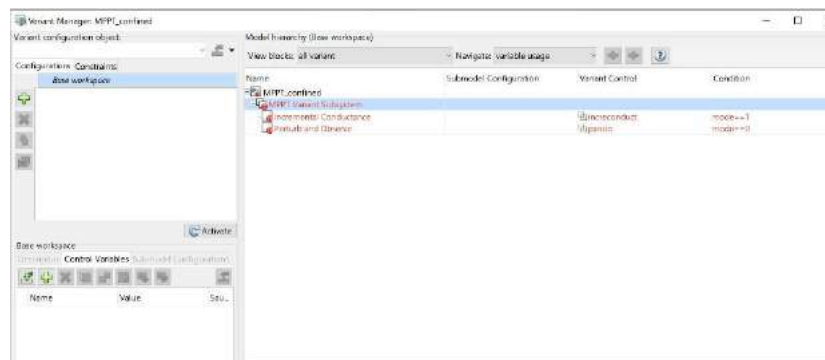
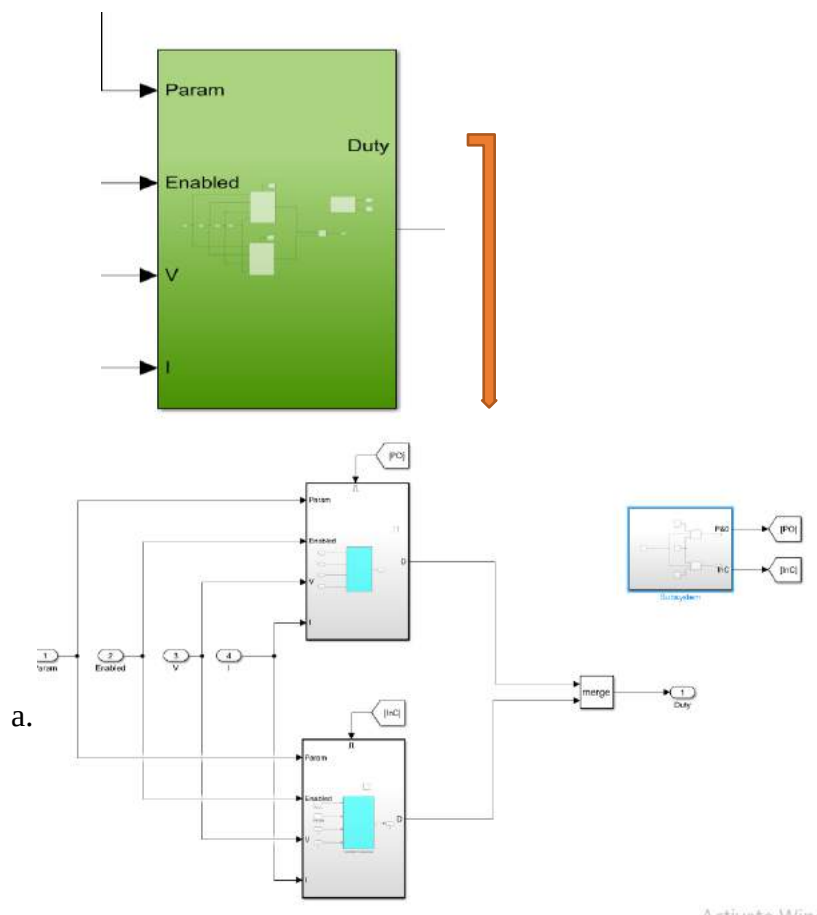


Figure 5.13: Variant Manager

5.4.5.2 Triggered Subsystem

To make this method work, a switch block is used to integrate an additional subsystem that describes the logic for switching between strategies.

The slider switch modifies the value of a constant block called PV mode, which serves as an input to the Simulink Signal Routing Library's switch block.



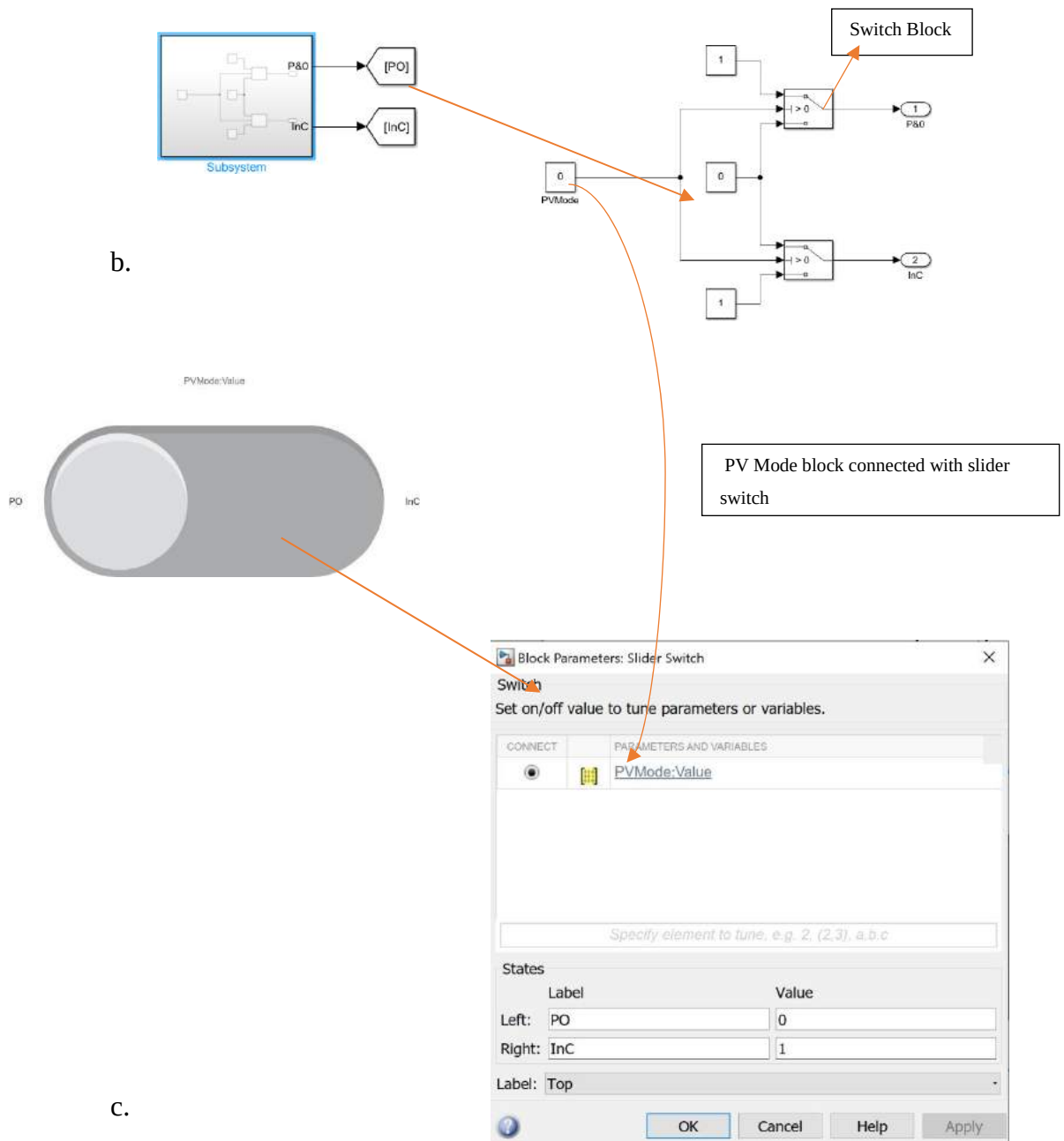
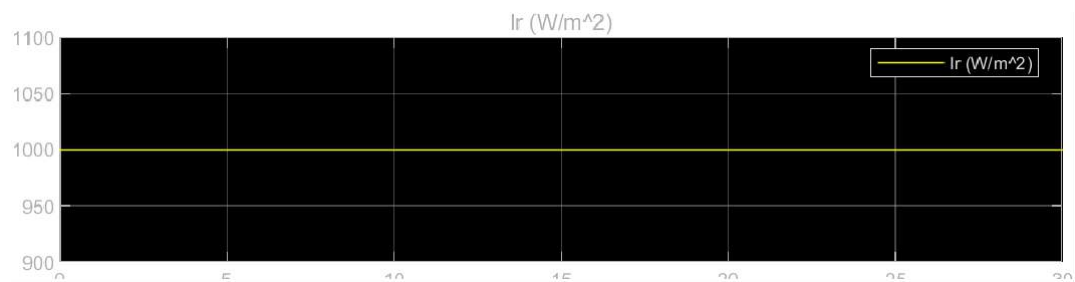


Figure 5.14: a.- Expanded View of the Triggered Subsystem, b.- Logic that triggers the subsystem, c.- Block Parameter box for Slider Switch.

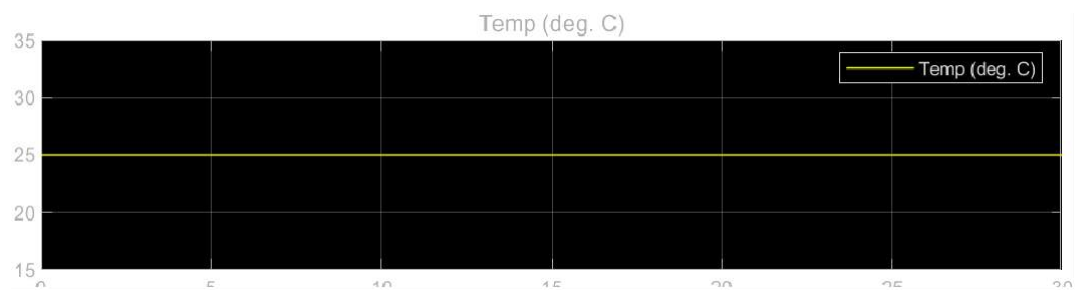
6. SIMULATION AND RESULTS

The simulation results obtained were very much like the literature review. The Figure 6.1 and 6.2 shows the waveforms obtained from the Monitoring block in the circuit model.

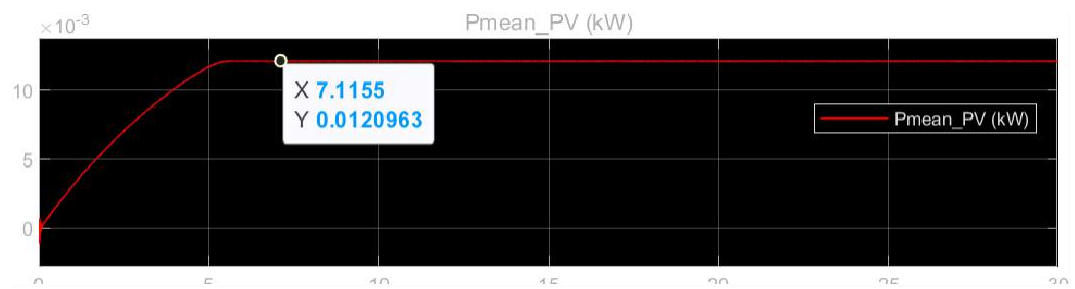
Case 1: Constant temperature @25 and Irradiation



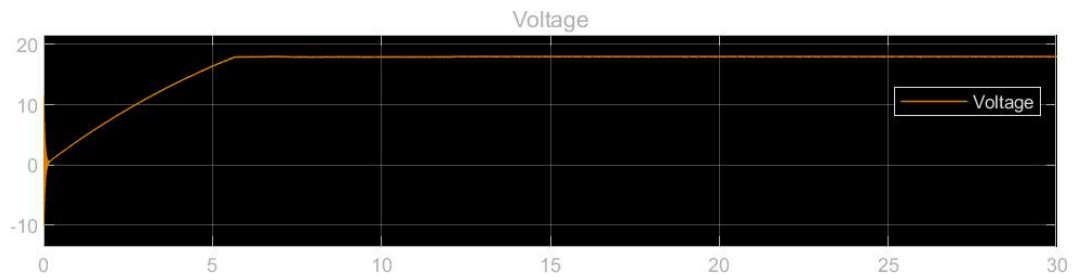
a.



b.



c.



d.

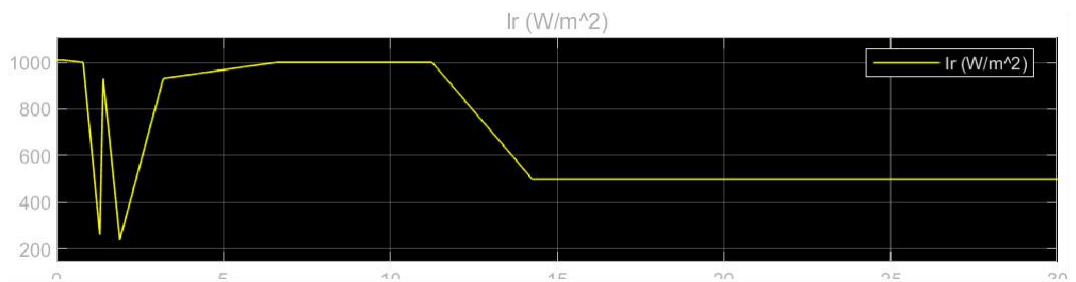


e.

Figure 6.1 a. Irradiance (W/m^2) b. Temperature (Celsius) c. Mean Power of PV array(kW)
d. PV_Voltage (V) e. Duty Cycle at Constant Irridance

The results obtained from the both the algorithms is same for constant irradiance.

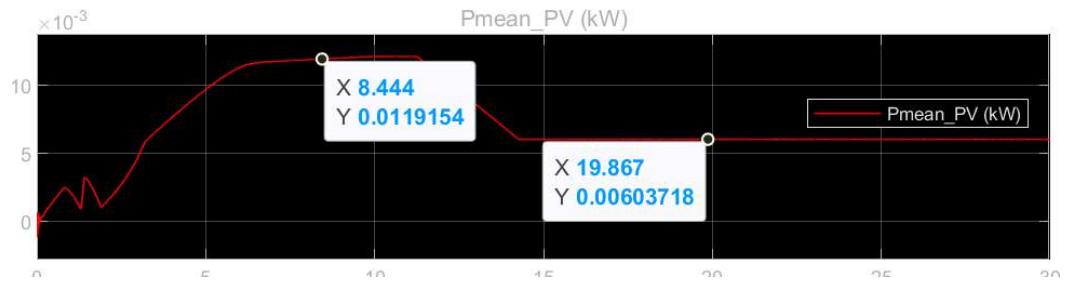
Case 2: Rapid Changing Temperature and Insolation



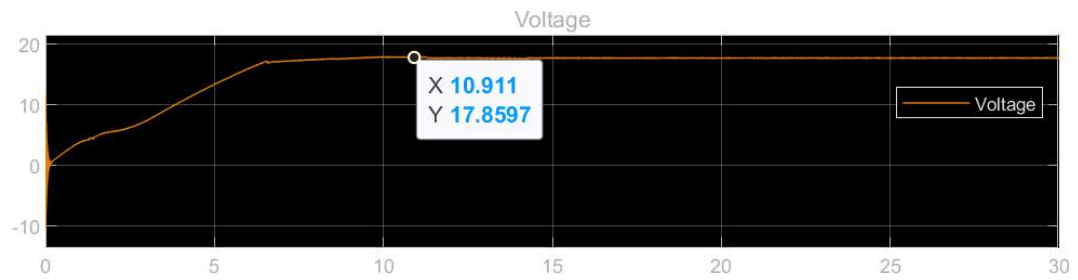
a.



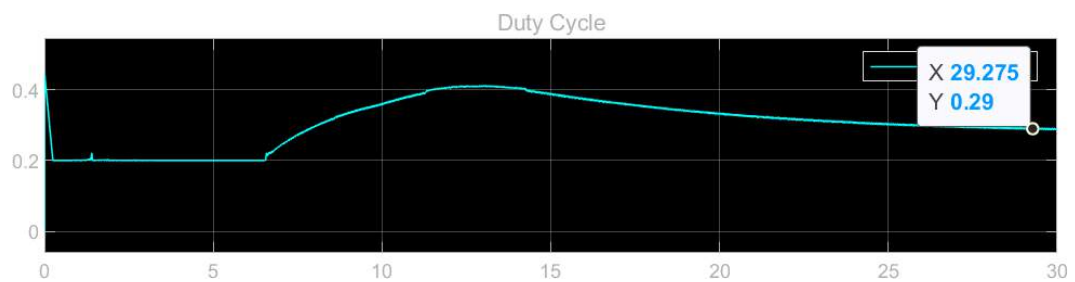
b.



c.



d.



e.

Figure 6.2 a. Irradiance (W/m^2) b. Temperature (Celsius) c. Mean Power of PV array(kW)
d. PV_Voltage (V) e. Duty Cycle

The results obtained in both the cases match the values to the PV array as seen in Figure 6.3 and we observe that due to the MPPT algorithms the whole module operates on the MPP once the point is attained.

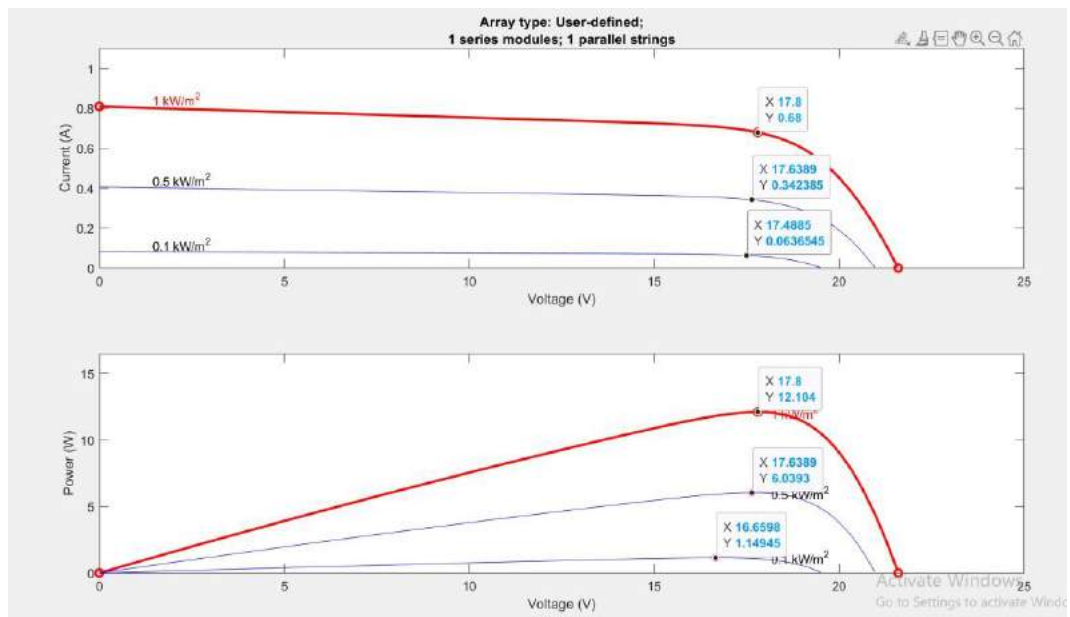


Figure 6.3 PV Array Characteristics

When we examine the signal in data inspector under varying irradiance conditions, we notice that the waveforms acquired from Perturb and Observe and Incremental Conductance are not the same, and there is a discrepancy between them as shown in Figure 6.4.

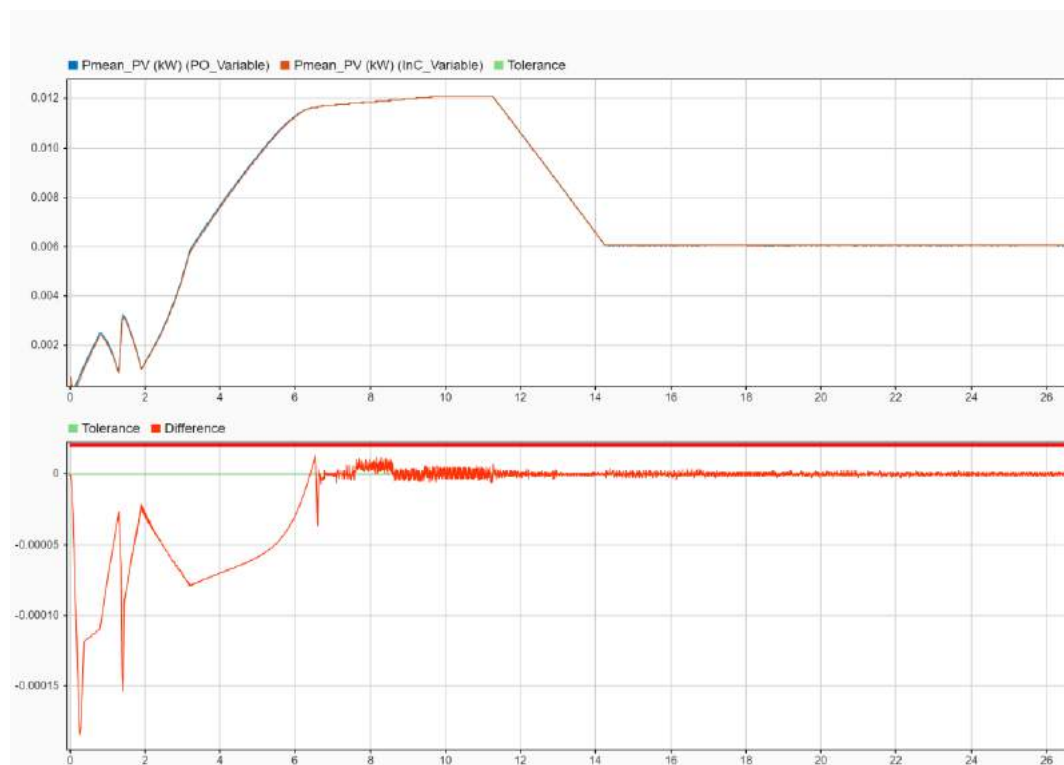


Figure 6.4 Comparison on Data Inspector Tool

The second plot in Figure 6.4 indicates the differences between both the techniques.

The steady state oscillations can also be observed from the data inspector as seen in Figure 6.5.

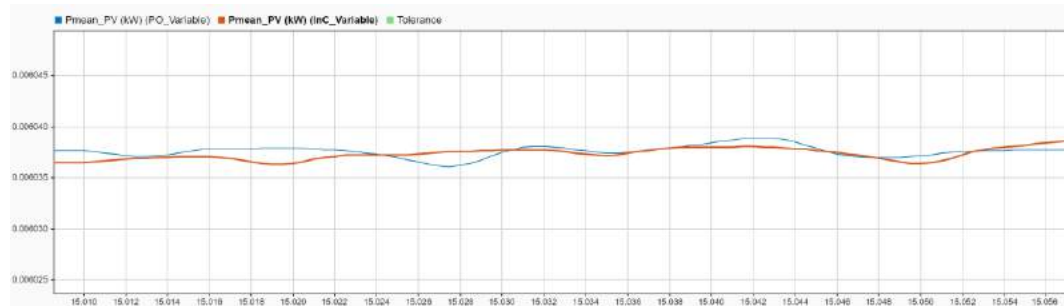


Figure 6.5 Steady State Oscillations

7. CONCLUSION

We learned about the importance of renewable energy in this project, and we looked into how we might best utilize this energy in terms of solar power.

The traditional Perturb and Observe and Incremental Conductance MPPT algorithms were simulated on MATLAB, and the results were satisfactory and in line with the literature reviews. Though the algorithms in the project had flaws, they were nevertheless capable of harvesting 94-96 percent of the solar energy incident on the solar modules. Newer technologies are continually being researched, and we expect that Solar Energy will become cost effective and extremely efficient in the near future.

We have also compared the Chopper circuits that are used to connect with the MPPT and come to some conclusions about which is the best option.

We investigated the Simulink block in depth and were successful in becoming acquainted with the MATLAB platform.

References

1. J. Mahanta, B. J. Saharia and N. Sarmah, "Comparative Performance Analysis of Chopper circuits in MPPT for Solar PV applications," 2018 International Conference on Current Trends towards Converging Technologies (ICCTCT), 2018, pp. 1-6, doi: 10.1109/ICCTCT.2018.8550940.
2. S. Gaikwad and M. Ghosal, "Energy efficient storage-less and converter-less renewable energy harvesting system using MPPT," 2017 2nd International Conference for Convergence in Technology (I2CT), 2017, pp. 971-973, doi: 10.1109/I2CT.2017.8226273.
3. Meskani, A., Haddi, A., Becherif, M., 2015. Modeling and simulation of a hybrid energy source based on solar energy and battery. *Int. J. Hydrogen Energy* 40, 13702–13707
4. Muhammad Kamran, Muhammad Mudassar, Muhammad Rayyan Fazal, Muhammad Usman Asghar, Muhammad Bilal, Rohail Asghar, Implementation of improved Perturb & Observe MPPT technique with confined search space for standalone photovoltaic system, *Journal of King Saud University - Engineering Sciences*, Volume 32, Issue 7, 2020, Pages 432-441, ISSN 1018-3639,
5. Raghavendra, K.V.G.; Zeb, K.; Muthusamy, A.; Krishna, T.N.V.; Kumar, S.V.S.V.P.; Kim, D.-H.; Kim, M.-S.; Cho, H.-G.; Kim, H.-J. A Comprehensive Review of DC–DC Converter Topologies and Modulation Strategies with Recent Advances in Solar Photovoltaic Systems. *Electronics* **2020**, *9*, 31. <https://doi.org/10.3390/electronics9010031>
6. M. Zerouali, S. Zouirech, A. E. Ougli, B. Tidhaf and H. Zrouri, "Improvement of Conventional MPPT Techniques P&O and INC by Integration of Fuzzy Logic," 2019 7th International Renewable and Sustainable Energy Conference (IRSEC), 2019, pp. 1-6, doi: 10.1109/IRSEC48032.2019.9078330.
7. S. Bhattacharyya, D. S. Kumar P, S. Samanta and S. Mishra, "Steady Output and Fast Tracking MPPT (SOFT-MPPT) for P&O and InC Algorithms," in *IEEE Transactions on Sustainable Energy*, vol. 12, no. 1, pp. 293-302, Jan. 2021, doi: 10.1109/TSTE.2020.2991768.
8. Baharudin, Nor & Tunku Mansur, Tunku Muhammad Nizar & Hamid, Fairuz & Ali, Rosnazri & Irwanto, Muhammad. (2017). Topologies of DC-DC converter in solar PV applications. *Indonesian Journal of Electrical Engineering and Computer Science*. 8. 368-374. 10.11591/ijeecs.v8.i2.pp368-374.
9. S. Zouirech, M. Zerouali, H. Elaisaoui, A. E. Ougli and B. Tidhaf, "Application of Various Classical and Intelligent MPPT Tracking Techniques for the

Production of Energy through a Photovoltaic System," 2019 7th International Renewable and Sustainable Energy Conference (IRSEC), 2019, pp. 1-6, doi: 10.1109/IRSEC48032.2019.9078154.

10. M. A. Elgendy, B. Zahawi and D. J. Atkinson, "Assessment of Perturb and Observe MPPT Algorithm Implementation Techniques for PV Pumping Applications," in IEEE Transactions on Sustainable Energy, vol. 3, no. 1, pp. 21-33, Jan. 2012, doi: 10.1109/TSTE.2011.2168245.

11. R. Arulmurugan*, N. Suthanthira Vanitha "Optimal Design of DC to DC Boost Converter with Closed Loop Control PID Mechanism for High Voltage Photovoltaic Application" International Journal of Power Electronics and Drive System (IJPEDS) Vol.2, No.4, December 2012, pp. 434~444 ISSN: 2088-8694.

12. A.A. A. Hafez. Multi-level cascaded DC/DC converters for PV applications. Alexandria Eng. J. 2015; 54(4): 1135–1146.

13. M. Orabi, F. Hilmy, A. Shawky, J. A. Abu Qahouq, E. S. Hasaneen, E. Gomaa. On-chip integrated power management MPPT controller utilizing cell-level architecture for PV solar system. Sol. Energy. 2015; 117: 10–28.

14. [dc-to-dc-converter-in-maximum-power-pointtracker.pdf \(rroj.com\)](#)

15. <https://www.irjet.net/archives/V6/i6/IRJET-V6I6422.pdf>

16. <https://www.nptel.ac.in/noc/courses/noc20/SEM2/noc20-ee57/>